# Scalability & Future Plan

## SMARTBRIDGE - Scalability & Future Plan

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 1 Mark

## Current System Limitations

| S.No | Limitation | Impact | Priority to Address (High / Medium / Low) |
|---|---|---|---|
| **1** | **Local Server Runtime** | API is hosted on local Node.js engine (localhost:5000), restricting public internet accessibility. | **High** |
| **2** | **External AI API Rate Limits** | Heavy concurrent calls to Gemini AI may hit external API quota limits. | **Medium** |
| **3** | **No Visual Frontend UI** | Project currently operates as a REST API consumed via Postman / Thunder Client. | **Medium** |

## Scalability Plan

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| **1** | **User Load** | Single-instance Express Node.js server. | Deploy to Render / AWS ECS with PM2 cluster mode and auto-scaling. |

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|---|---|---|---|
| **2** | **Data Storage** | MongoDB Atlas M0 Free Tier Cluster. | Upgrade to MongoDB Atlas Dedicated M10 Cluster with indexing on user and workoutDate. |
| **3** | **Performance** | Direct call to Gemini AI for every recommendation. | Implement Redis caching layer for repeated workout plan queries to reduce latency. |
| **4** | **Security** | Standard JWT Auth & Bcrypt Hashing. | Implement Express-rate-limit, CORS origin restriction, and Refresh Token rotation. |

## Future Roadmap

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|
| **Phase 1** | **Cloud Deployment & CI/CD** | Q4 2026 | Deploy backend API to Render/AWS with automated GitHub Actions testing pipeline. |
| **Phase 2** | **Frontend Web Dashboard** | Q1 2027 | Build React.js frontend interface with charts for workout statistics and AI recommendations. |
| **Phase 3** | **Wearable Device Integration** | Q2 2027 | Sync real-time step and heart rate data from Apple HealthKit / Google Fit APIs. |
| **Phase 4** | **Multimodal Vision AI** | Q3 2027 | Use Gemini Vision API to analyze food images and auto-calculate meal calories. |